Preliminary Round Submission

Claim IQ
Decision Intelligence Platform for Malaysian TPA Claims Adjudication

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026
umhackathon@um.edu.my

GROUP: AmaChuuErs
Domain: Domain 2 - AI for Economic Empowerment & Decision Intelligence
Submitted: 26th April 2026

Table of Contents

Target Domain....4
Proposed Solution Summary....4
2. Background & Business Objective....4
3. Product Purpose....5
3.1. Main Goals of the System....5
3.2. Intended Users (Target Audience)....5
3.3. Success Metrics....5
4. System Functionalities....7
4.1. Description....7
4.2. Core Functionalities....7
4.2.1. System Architecture Diagram....7
4.3. AI Model & Prompt Design....9
4.3.1. Model Selection....9
4.3.2. Prompting Strategy....9
4.3.3. Context & Input Handling....9
4.3.4. Fallback & Failure Behaviour:....9
User Stories & Use Cases....9
Use Cases (Primary Interactions)....9
Use Case Diagram:....11
Features Included (Scope Definition)....11
Features Not Included (Scope Control)....11
Assumptions & Constraints....11
Risks & Questions Throughout Development....12

References.....13

Project Name: ClaimIQ
Version: 2.0

ClaimIQ helps third-party administrators (TPAs), clinic teams, and medical reviewers turn a messy outpatient claim into a clear, explainable decision packet before reimbursement gets delayed. That matters because weak claims operations create real cost and trust problems: the AMA reported in 2025 that practices complete 39 prior authorization requests per physician per week, spend about 13 hours weekly on that work, and often dedicate staff just to managing it [1]. While prior authorization is not identical to claim adjudication, it shows the same kind of administrative burden ClaimIQ is designed to reduce.

Target Domain

AI for Economic Empowerment & Decision Intelligence in Healthcare Claims Operation, Medical Cost Containment, and Provider Reimbursements.

Proposed Solution Summary

ClaimIQ is a decision-intelligence copilot for outpatient claims. It is designed first for TPA operators, who often need to make fast decisions with incomplete evidence, and it also supports clinics, medical reviewers, fraud investigators, and operations managers. Using Z.AI GLM as its core reasoning layer, the platform turns medical evidence into structured claim data, pulls the right policy clauses, flags inconsistency or fraud risk, drafts a clear rationale, and sends uncertain cases to manual review instead of forcing unsafe automation [5].

2. Background & Business Objective

BACKGROUND OF THE PROBLEM: Outpatient claim review is still slowed down by scattered evidence, manual policy checks, and inconsistent decision records. The wider utilization-management workflow shows how heavy that burden has become: the AMA reported in 2025 that physicians complete an average of 39 prior authorization requests per week, spend about 13 hours each week on that work, and 40% employ staff dedicated to it [1]. On the leakage side, the NHCAA estimates healthcare fraud losses at roughly 3% to 10% of total health spending, which shows why stronger detection and documentation matter before payment is released [2]. In Malaysia, medical claims friction is visible enough that PIAM highlighted the need for stronger collaboration and transparency across insurers, TPAs, and healthcare partners in January 2026 [3].

Importance of Solving this Issue: Faster, better-explained claim decisions have direct economic value. Clinics get reimbursed sooner, TPA teams spend less time chasing missing fields or reworking denials, medical reviewers can focus on the cases that actually need judgment, and payers can reduce leakage from miscoding, ineligible claims, and suspicious billing patterns. In Malaysia, pressure around TPA-clinic relationships has also become part of the story: CodeBlue published a 2025 commentary alleging delayed settlements, fee deductions, and 'double-dipping' practices affecting GPs [4]. We use that article here as evidence of ecosystem tension, not as a settled regulatory finding.

Strategic Fit / Impact: ClaimIQ fits Domain 2 because it improves economic outcomes through better operational decisions, not just faster paperwork. Z.AI GLM sits at the center of that value because the product depends on grounded reasoning, explainable recommendations, and usable natural-language outputs across coding support, adjudication, fraud narratives, appeal drafting, claim chat, and weekly intelligence reports [5]. The intended result is shorter turnaround time, lower review cost per claim, and earlier detection of anomalies before payment is released.

3. Product Purpose

3.1. Main Goals of the System: To help TPA operators and clinic teams make faster, safer, and more auditable reimbursement decisions by combining deterministic validation, Z.AI GLM reasoning, and human-review safety gates.

3.2. Intended Users (Target Audience)

• Primary user: TPA (third-party administrator) operators who review intake quality, policy fit, and decision recommendations at scale

• General practitioners and clinic administrators submitting outpatient claims and supporting evidence

• Medical reviewers who require explainable clinical reasoning before approving, adjusting, or escalating a case

• Fraud investigators and auditors monitoring anomaly patterns across claims, providers, and clinics

• Insurance operations managers tracking denial rates, turnaround time, clinic performance, and intervention priorities

3.3. Success Metrics

- Reduce median claim-preparation time by at least 50% on the hackathon benchmark set
- Keep autonomous denials at zero for low-confidence or incomplete claims; those cases must route to manual review

- Generate a reviewer-ready decision packet in under two minutes for standard outpatient claims
- Catch at least 90% of seeded fraud or anomaly scenarios in the demo dataset
- Improve explanation quality so a reviewer can trace every recommendation back to evidence, policy, or a deterministic rule

4. System Functionalities

4.1. Description: ClaimIQ is an end-to-end claims intelligence workflow. It takes in clinical notes and supporting documents, runs deterministic intake validation and eligibility checks, turns evidence into structured fields, adds policy context, and then uses Z.AI GLM to generate coding support, adjudication rationale, fraud analysis, appeal support, and explanations. The product experience includes an operator dashboard, claim detail views, analytics, and a GP-facing interaction flow. In the current implementation, those capabilities are backed by FastAPI endpoints, SQLite persistence, FAISS-based retrieval, reviewer routing logic, and bilingual advisory and appeal outputs **[5].**

4.2. Core Functionalities: Claim intake and scrubbing check identity format, missing fields, unusual amounts, and whether the submission is ready for downstream reasoning. Eligibility verification checks coverage dates, visit validity, visit limits, deductible or copay conditions, and remaining outpatient balance so clearly ineligible claims are caught early. Multi-modal evidence parsing uses document triage and supporting vision models to turn loose attachments into structured fields that can be checked against doctor notes. Policy retrieval plus Z.AI GLM reasoning then suggest ICD-10 or CPT-style codes, benefit logic, and adjudication rationale tied back to retrieved clauses. Fraud and clinical consistency checks look for likely upcoding, non-covered care, unusual charges, and contradictions between objective evidence and subjective notes. Human-in-the-loop review, appeal drafting, and analytics complete the workflow with decision packets, claim chat, weekly reporting and KPI visibility, and explanation-of-benefits outputs.

4.2.1. System Architecture Diagram

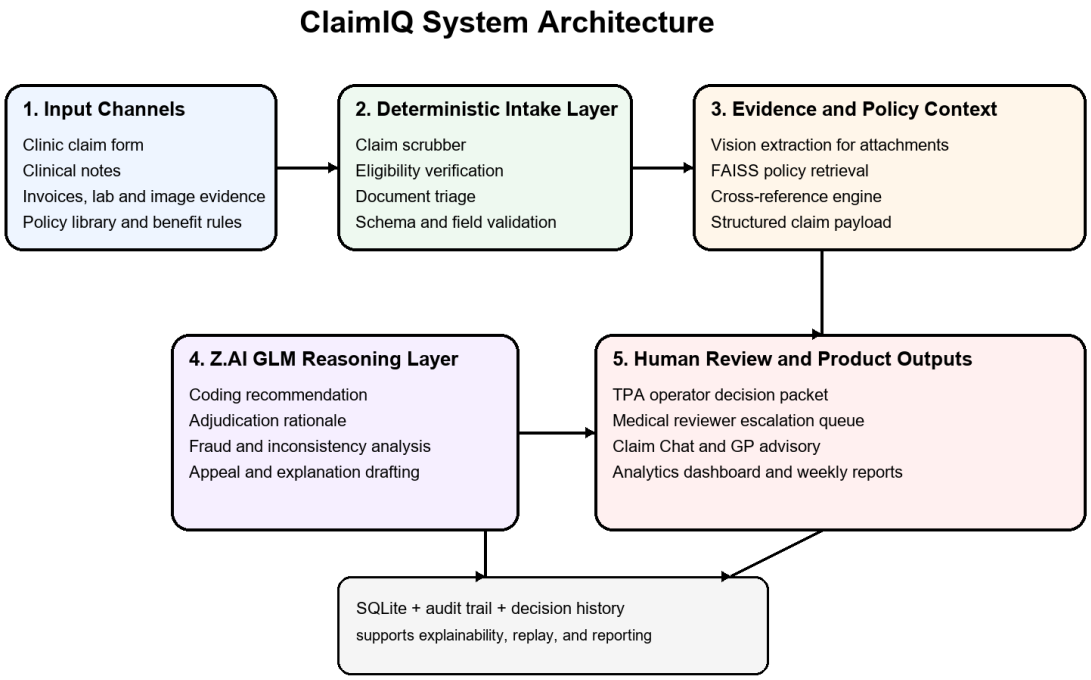

Figure: ClaimIQ combines deterministic validation with Z.AI GLM reasoning and human review safety gates

ClaimIQ system architecture showing the deterministic, reasoning, and human-review layers

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI GLM is the central reasoning model for ClaimIQ because the workflow needs four things at once: strong policy and clinical reasoning over mixed structured and unstructured inputs, reliable JSON-style outputs for downstream orchestration, explanations that human reviewers can trust, and repeatable behavior across multi-step prompts. Supporting vision models are used only to extract attachment content; final decision support always flows back through deterministic checks plus Z.AI GLM [5].

4.3.2. Prompting Strategy: ClaimIQ uses a staged prompting strategy instead of one large prompt. Separate steps handle extraction, coding support, adjudication, fraud analysis, GP advisory, and appeal drafting, each with its own instructions and output contract. That makes the workflow easier to validate, easier to debug, and safer to control because only the fields needed for the next step are carried forward.

4.3.3. Context & Input Handling: Raw clinical narrative is validated at intake, along with patient identifiers, visit date, and required financial fields, before model calls begin. Large policy documents are not pushed directly into the prompt; they are chunked, embedded, and stored in a local FAISS index so only the most relevant clauses are retrieved into Z.AI GLM context. Oversized or noisy evidence is processed into structured summaries where possible, and poor-quality inputs are either rejected early or routed for manual review instead of being passed forward blindly.

4.3.4. Fallback & Failure Behaviour: ClaimIQ is fail-safe. Z.AI GLM responses are retried, markdown fences are stripped before JSON parsing, and malformed outputs are never accepted silently. If the Z.AI GLM service is unavailable or returns unusable output, the claim is not denied automatically; it is saved with the completed deterministic checks and routed to manual review with a visible failure reason. Fraud gates and safety gates can also override a model recommendation whenever evidence quality, risk level, or policy ambiguity is high [5].

User Stories & Use Cases

User Stories: As a TPA operator, I want the relevant evidence, policy clauses, coding support, and risk signals gathered into one decision packet so I can review more claims without jumping between systems. As a GP, I want to submit notes and supporting documents once and receive a recommendation that is understandable and defensible. As an operations manager, I want weekly KPI and denial intelligence so I can spot leakage, bottlenecks, and clinics that need intervention.

Use Cases (Primary Interactions)

Claim Submission and Review Flow: A clinic submits patient details, visit date, amount, and clinical notes, and may attach invoices or supporting evidence. ClaimIQ performs scrubbing and eligibility checks, extracts structured evidence, retrieves the relevant policy context, asks Z.AI GLM for coding

and adjudication support, estimates fraud risk, and returns a structured decision output for operator review.

Fraud Escalation Flow: A fraud reviewer opens a flagged claim, reviews the evidence gaps and risk indicators, uses Claim Chat for follow-up questions, and decides whether the case should be escalated before payout.

Appeal Support Flow: If a claim is disputed, the operator provides the appeal reason and any new supporting evidence. ClaimIQ then uses Z.AI GLM to draft a rebuttal grounded in the original denial context, retrieved policy evidence, and the updated claim data.

Portfolio Intelligence Flow: Managers review clinic-level analytics, denial-code breakdowns, fraud trends, and weekly narrative summaries to decide where policy communication, audit effort, or staffing changes are most needed.

Use Case Diagram for ClaimIQ

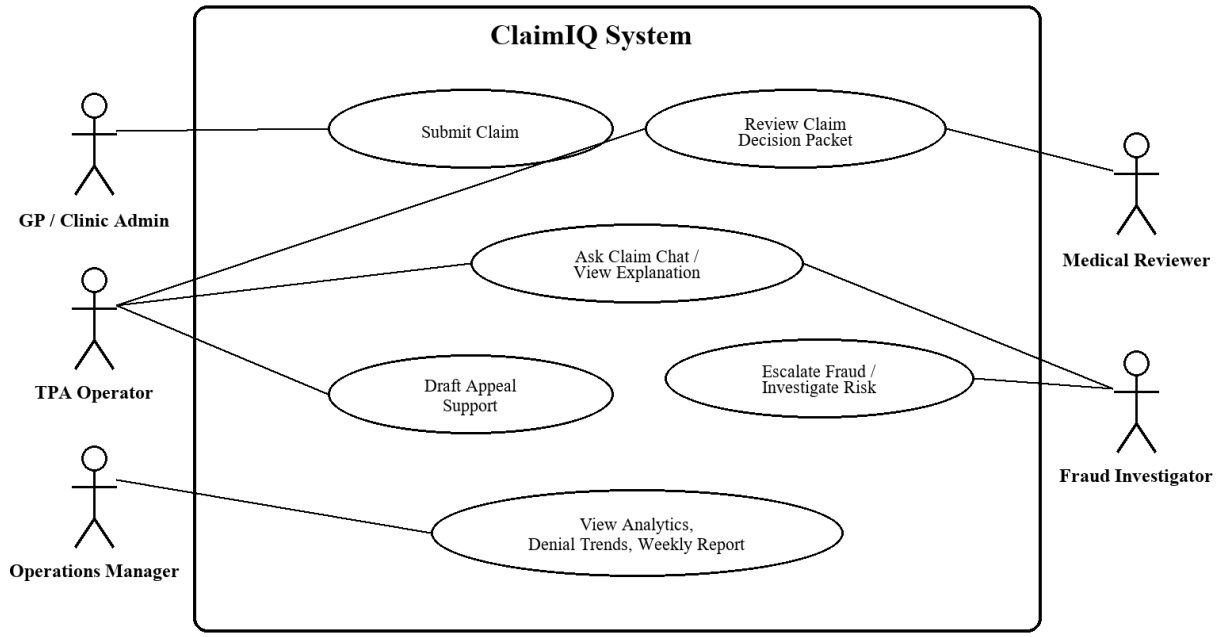

Figure: ClaimIQ use case diagram showing the main actors and interactions

Use Case Diagram: The diagram below shows the main ClaimIQ actors and the product interactions each role relies on during claim submission, review, escalation, appeals, and analytics.

Features Included (Scope Definition)

- Core intake workflow: FastAPI claim submission, claim scrubbing, and eligibility validation before reasoning begins [5]
- Evidence understanding: document triage and extraction for clinical notes, invoices, lab reports, and imaging attachments [5]
- Policy intelligence: FAISS-backed retrieval of relevant benefit clauses, limits, and coding context [5]
- Z.AI GLM decision support: coding suggestions, adjudication rationale, fraud analysis, GP advisory, and appeal drafting [5]
- Human-review safety layer: manual-review routing for ambiguous, high-risk, or low-confidence claims [5]
- Operational tooling: claim queue, Claim Chat, explanation-of-benefits output, and reviewer decision packets [5]
- Management visibility: denial analytics, fraud heatmaps, clinic performance views, and weekly narrative reporting [5]
- System recordkeeping: SQLite-backed audit trail, claim history, fraud scores, and scenario-based demo validation data [5]

Features Not Included (Scope Control)

- Direct insurer-core or hospital EMR integration in the current hackathon version
- Real payment settlement, remittance, or banking rails
- A production-ready patient mobile application
- Fully autonomous final denial without any human sign-off
- Enterprise-scale ingestion for every file type and every quality of scanned medical evidence

Assumptions & Constraints

LLM Cost Constraint: On a per-claim basis, estimated model spend is kept low through inference-light claim validation, fetching only the most relevant chunks from policies instead of entire policy manuals, and separating expensive reasoning from deterministic checks such as eligibility and scrub rules. The aim is to balance inference cost with decision quality.

Technical Constraints: The current workflow relies on valid API keys, a built FAISS policy index, and reasonably clear evidence images. If uploads are blurry, cropped, or incomplete, document understanding gets worse, and some vision-extraction pathways also depend on external model endpoints. The implementation currently uses SQLite in the working system, even though the architecture could later be extended to larger database deployments [5].

Performance Constraints: End-to-end speed depends on how much evidence is attached, whether the system is starting cold, and how quickly external models respond. Deterministic steps are fast, but vision extraction and Z.AI GLM reasoning can take several seconds per claim depending on evidence volume and model latency. The system favors safe review quality over instant turnaround.

The product assumes that the clinic or operator submits a minimally complete claim, including valid patient identifiers, the visit date, and a clinically meaningful narrative. Claims with missing or low-quality evidence can still be reviewed, but the confidence level and depth of automation will be lower.

Validation Assumption: For the hackathon build, scenario-based and synthetic claim datasets are acceptable stand-ins for production insurer traffic, as long as they reflect realistic edge cases and show measurable decision-support value.

Risks & Questions Throughout Development

Clinical and Policy Accuracy – How do we put continuous checks in place so coding and adjudication outputs do not drift as payer rules change?

False Positives in Fraud Detection – How do we avoid over-escalating legitimate claims from clinics whose case mix and pricing patterns are unusual but not actually fraudulent?

Data Privacy and Governance: What additional controls are required before handling real production medical data under stricter compliance expectations?

Operational Adoption: What level of explanation, confidence scoring, and workflow integration is needed for reviewers to trust the recommendations and use them consistently?

Model and Dependency Resilience: What should the system do when Z.AI GLM, MedGemma, Gemini, or the retrieval layer is slow or unavailable during a peak processing window?

Testing and Rollout: What extra scenario coverage, benchmark datasets, and human-review metrics are needed to move from a hackathon prototype to a live pilot?

References

[1] American Medical Association. 'Fixing prior auth: Nearly 40 prior authorizations a week is way too many.' 24 April 2025. https://www.ama-assn.org/practice-management/prior-authorization/fixing-prior-auth-nearly-40-prior-authorizations-week-way

[2] National Health Care Anti-Fraud Association. 'The Challenge of Health Care Fraud.' Accessed April 25, 2026. https://www.nhcaa.org/tools-insights/about-health-care-fraud/the-challenge-of-health-care-fraud/

[3] PIAM. 'Healthcare Partners Protocol & Solutions Committee (HPPSC) Strengthens Collaboration to Enhance Medical Claims Management.' 23 January 2026. https://piam.org.my/news-media/stay-ahead/press-releases/article/Healthcare-Partners-Protocol---Solutions-Committee-HPPSC-Strengthens-Collaboration-to-Enhance-Medical-Claims-Management/

[4] CodeBlue. 'The Double-Dipping Scandal: How TPAs Are Exploiting Both Corporations And Clinics — Dr James Jeremiah.' 13 November 2025. Used as a published industry allegation/opinion, not as a regulatory or audited fact. https://codeblue.galencentre.org/2025/11/the-double-dipping-scandal-how-tpas-are-exploiting-both-corporations-and-clinics-dr-james-jeremiah/

[5] ClaimIQ codebase evidence from execution/api_server.py, execution/claims_processor.py, execution/database.py, execution/rag_engine.py, and execution/frontend/ for product behavior, safety routing, bilingual outputs, FastAPI endpoints, SQLite persistence, and FAISS-backed retrieval.